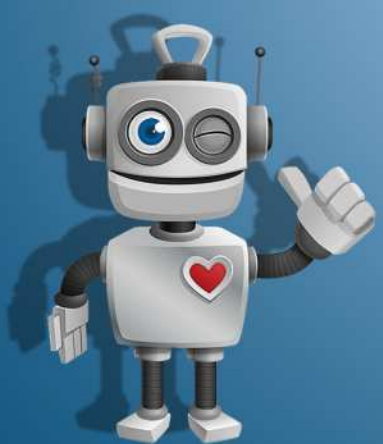
تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

امتحان سابق لمادة

التحكم بالعمليات





سكند



www.spectrum-fet.weebly.com

groups/mechatronics.groups

0785060940

Al-Balqa' Applied University
Faculty Of Engineering Technology
Mechatronics Engineering Dept.

Instructor ;Eng Monther Kan"an

Process Control
(3 Class Hours 20 Minutes.)
Tuesday-7—3-2016 (10-11)

second
Exam'08'
20 Points

2nd Semester
9:30 → 11

Student Name: .....

Q.1: Choose the correct answers ... {8 Points}

1. Two first –order systems with different time lags connected in cascade will produce ..... Second –order response.

a. overdamped b. undrdamped c. criticalydamped

2- For under damped second order controlled system, the damping ratio is .....

a. Greater than 1 b. equal 1 c. less than 1

3- The unit of 'time constant' of a system is the same as that of –

a. velocity b. time c. $(time)^{-1}$ d. non of above

4- Manipulated variable is the ..... That acted on by the controller .

a-control variable b- controlled variable

c- desired value d- process variable

5--If the error becomes more positive as the control variable decreases the action of controller is called.....

a- Direct Acting b- Reverse Acting

6- ..... Is an allowable fluctuation error

a- Dead time b- hysteresis c- dead band d- (b+c)

7- The objective of control system is to make the process variable zero

(a)True (b) False

8- The best control is larger duration with smaller peak error or vice versa.....

(a)True (b) False

1	2	3	4	5	6	7	8
A	C	B	B	B	D	B	A

Q2: (12 points)

: A process with a level a desired value of 150 mm head has a process variable input of 100 mm head

1- Express the error as percentage of range given that process variable has a range of a- 50mm to 150 mm b- 100mm to 1100mm

2- What is the magnitude and the negative sign indicates that?

SP = 150 mm PV = 100 mm

[1]

a) (50 − 150) mm

$E\% = \frac{SP - PV}{PV_{max} - PV_{min}} = \frac{(150 - 100)\,mm}{(150 - 50)\,mm}$

$= -0.5$

$E\% = -50\%$

b) (100 − 1100) mm

$E\% = \frac{SP - PV}{PV_{max} - PV_{min}} = \frac{150 - 100}{100 - 1100}$

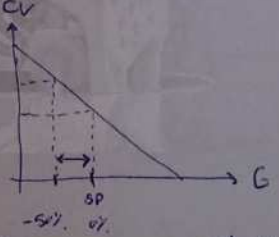
$= -0.05$

$E\% = -5\%$

[2]

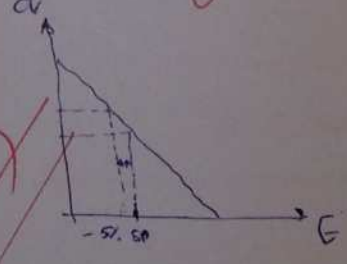
هذه الاشاره تعني انه احتاج الى فتح (تشغيل) ال final elemnt حتى أصل الى ال setpoint 50%

If heating system



في مثل هذه الحالة يجب علينا تقليل ال CV للتقليل ال E

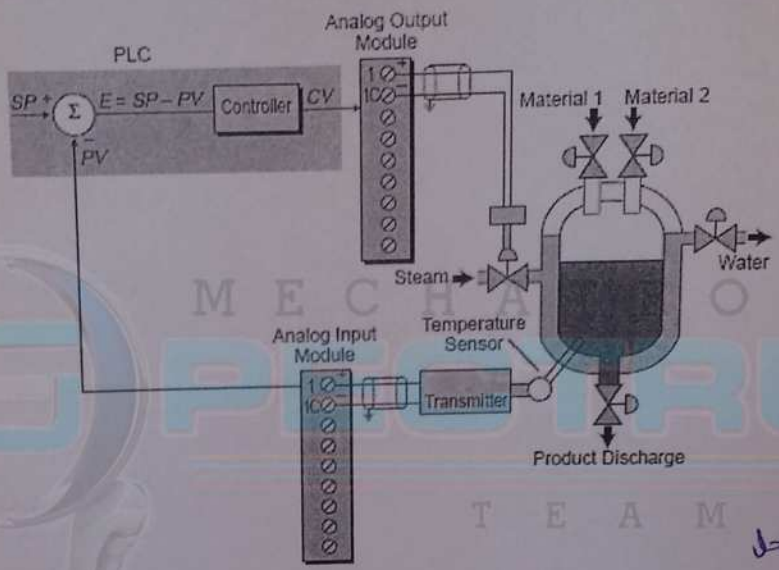
هذه الاشاره تعني أنه أحتاج الى تشغيل ال final elemnt بمقدار 5% حتى أصل الى ال setpoint



نلاحظ من خلال هذه المنحنيات أنه كلما زاد ال PV. Range كلما قلت نسبة الخطأ

⇐ الحل باللغة الانجليزية.

B- for this system below plot A control loop for it ?



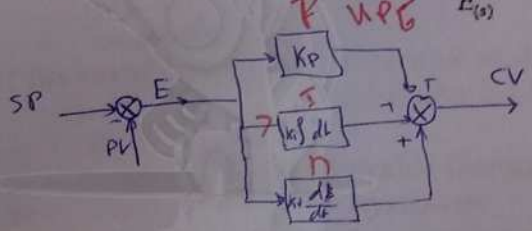
اتبع الحل ⟹

Control system block diagram including I/O interfaces.

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b- Prove that the transfer function of a PID controller (Hc)in laplace is

$$\frac{CV_{(s)}}{E_{(s)}} = K_P + \frac{K_I}{s} + K_D s$$



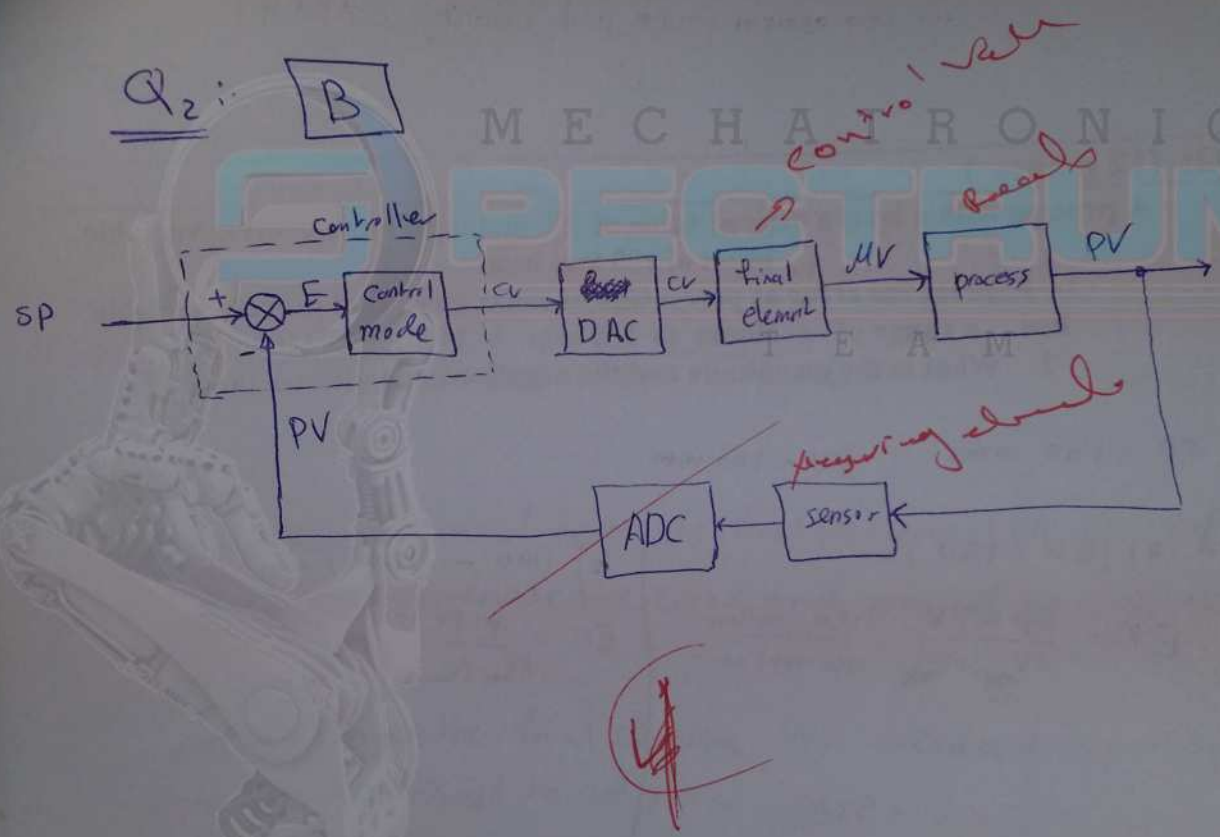
~~$CV(t) = K_P E(t) + K_i \int E(t)\,dt + K_d \frac{d}{dt}$~~

$$CV(t) = K_P E(t) + K_i \int E(t)\,dt + K_d \frac{dE(t)}{dt} \leftarrow CV$$

$$CV(s) = E(s)\left[K_P + \frac{K_i}{s} + K_d s\right]$$

$$\frac{CV(s)}{E(s)} = K_P + \frac{K_i}{s} + K_d s \quad //$$

Q2 : B


Controller
SP
+
-
E
Control mode
CV
DAC
CV
final element
MV
process
PV
PV
ADC
sensor